

4,212,2 Macroeconomics III

Exercises and Independent Studies

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Outline

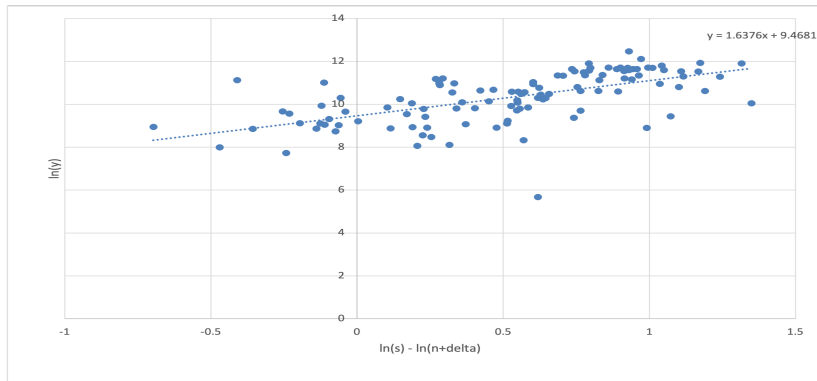
- ▶ The Solow Model with Human Capital (Exercise 3.1)
- ▶ The Solow Model with Oil (Exercise 3.2)

The Solow Model with Human Capital: Motivation

- ▶ Remember Exercise 1.3.
- ▶ The Solow model predicts that there should be a linear relationship between $\ln(y^*)$ and $\ln(s) - \ln(n + \delta)$, and that the slope should be $\frac{\alpha}{1-\alpha}$, which is equal to 0.5 if $\alpha = 1/3$:

$$\ln(y^*) = \frac{1}{1-\alpha} \ln(B) + \frac{\alpha}{1-\alpha} \ln(s) - \frac{\alpha}{1-\alpha} \ln(n + \delta).$$

The Solow Model with Human Capital: Motivation



The slope estimate is significantly larger than $1/2$ and we can conclude that in the real world, the structural parameters s and n seem to have a stronger influence on y than predicted by the theory.

The Solow Model with Human Capital: Motivation

- ▶ The slope estimate is $\frac{\alpha}{1-\alpha} = 1.63$, which implies that $\alpha = 0.62 \sim 2/3$.
- ▶ $\alpha = 2/3$ is too large to be attributable to physical capital.
- ▶ If we are confident that $\alpha = 1/3$, it follows that our OLS estimate is biased, meaning it is different than the 'true value'.
- ▶ Note: If we do not add all relevant variables in the empirical model, these will be left in the error term (residuals). And they could be correlated with s and n .
- ▶ Ideally, we should add something which is positively correlated with s , and negatively correlated with n .
- ▶ What about human capital (skills)?

The Solow Model with Human Capital: Motivation

- ▶ Human capital can be defined as the accumulated stock of what is invested in the training of the labor force: education, learning by doing, health, etc.
- ▶ Schooling tends to rise when fertility declines, hence it is negatively correlated with n .
- ▶ Countries with high investment rates in capital tend also to invest in schooling, hence human capital is positively correlated with s .
- ▶ Mass education and sustained growth took off around the same time (19th century).
- ▶ Thus, including human capital in the model seems worthwhile, as it might “solve” some of our problems.

The Solow Model with Human Capital

$$Y_t = (K_t)^\alpha (H_t)^\phi (A_t L_t)^{1-\alpha-\phi}, \quad (1)$$

$$K_{t+1} = I_t^K + (1 - \delta)K_t, \quad (2)$$

$$H_{t+1} = I_t^H + (1 - \delta)H_t, \quad (3)$$

$$w_t = (1 - \alpha) \left(\frac{K_t}{A_t L_t} \right)^\alpha \left(\frac{H_t}{A_t L_t} \right)^\phi A_t, \quad (4)$$

$$r_t = \alpha \left(\frac{K_t}{A_t L_t} \right)^{\alpha-1} \left(\frac{H_t}{A_t L_t} \right)^\phi, \quad (5)$$

$$C_t + S_t = Y_t, \quad (6)$$

$$S_t = I_t^K + I_t^H, \quad (7)$$

$$I_t^K = s_K Y_t, \quad (8)$$

$$I_t^H = s_H Y_t, \quad (9)$$

$$L_{t+1} = (1 + n)L_t, \quad (10)$$

$$A_{t+1} = (1 + g)A_t. \quad (11)$$

The Solow Model with Human Capital

$$Y_t = (K_t)^\alpha (H_t)^\phi (A_t L_t)^{1-\alpha-\phi}, \quad (1)$$

$$K_{t+1} = s_K Y_t + (1 - \delta) K_t, \quad (2)$$

$$H_{t+1} = s_H Y_t + (1 - \delta) H_t, \quad (3)$$

$$w_t = (1 - \alpha) \left(\frac{K_t}{A_t L_t} \right)^\alpha \left(\frac{H_t}{A_t L_t} \right)^\phi A_t, \quad (4)$$

$$r_t = \alpha \left(\frac{K_t}{A_t L_t} \right)^{\alpha-1} \left(\frac{H_t}{A_t L_t} \right)^\phi, \quad (5)$$

$$C_t + s_K Y_t + s_H Y_t = Y_t, \quad (6)$$

$$L_{t+1} = (1 + n) L_t, \quad (7)$$

$$A_{t+1} = (1 + g) A_t. \quad (8)$$

7 Parameters: g , α , s_K , s_H , n , ϕ and δ . 8 Endogenous variables: Y , K , L , r , w , C , A , and H of which K_t , H_t , A_t and L_t are state (predetermined) variables.

Transformations in per 'Efficiency Unit of Labor' Terms

Define: $\tilde{k}_t = \frac{k_t}{A_t} = \frac{K_t}{A_t L_t}$, $\tilde{y}_t = \frac{y_t}{A_t} = \frac{Y_t}{A_t L_t}$, $\tilde{h}_t = \frac{h_t}{A_t} = \frac{H_t}{A_t L_t}$. Then:

$$\frac{H_{t+1}}{A_{t+1} L_{t+1}} = s_H \frac{Y_t}{L_t(1+n)A_t(1+g)} + (1-\delta) \frac{H_t}{L_t(1+n)A_t(1+g)},$$

which is equivalent to

$$\tilde{h}_{t+1} = s_H \frac{\tilde{y}_t}{(1+n)(1+g)} + (1-\delta) \frac{\tilde{h}_t}{(1+n)(1+g)}.$$

Transformations in per 'Efficiency Unit of Labor' Terms

The set of equations with transformed variables in per efficiency unit of labor terms is:

$$\tilde{y}_t = (\tilde{k}_t)^\alpha (\tilde{h}_t)^\phi,$$

$$\tilde{k}_{t+1} = s_K \frac{\tilde{y}_t}{(1+n)(1+g)} + (1-\delta) \frac{\tilde{k}_t}{(1+n)(1+g)},$$

$$\tilde{h}_{t+1} = s_H \frac{\tilde{y}_t}{(1+n)(1+g)} + (1-\delta) \frac{\tilde{h}_t}{(1+n)(1+g)},$$

$$\tilde{w}_t = (1-\alpha)(\tilde{k}_t)^\alpha (\tilde{h}_t)^\phi,$$

$$r_t = \alpha(\tilde{k}_t)^{\alpha-1} (\tilde{h}_t)^\phi,$$

$$\tilde{c}_t = \tilde{y}_t(1 - s_K - s_H),$$

where $\tilde{w}_t = \frac{w_t}{A_t}$.

The Transition Equations

Replacing the production function: $\tilde{y}_t = (\tilde{k}_t)^\alpha (\tilde{h}_t)^\phi$ in the physical and human capital accumulation equations, we obtain the transition equations:

$$\tilde{k}_{t+1} = s_K \frac{(\tilde{k}_t)^\alpha (\tilde{h}_t)^\phi}{(1+n)(1+g)} + (1-\delta) \frac{\tilde{k}_t}{(1+n)(1+g)},$$

$$\tilde{h}_{t+1} = s_H \frac{(\tilde{k}_t)^\alpha (\tilde{h}_t)^\phi}{(1+n)(1+g)} + (1-\delta) \frac{\tilde{h}_t}{(1+n)(1+g)}.$$

The Solow Equations

Subtracting $\tilde{k}_t$ and $\tilde{h}_t$, respectively, on both sides of the transition equations gives the Solow equations:

$$\tilde{k}_{t+1} - \tilde{k}_t = s_K \frac{(\tilde{k}_t)^\alpha (\tilde{h}_t)^\phi}{(1+n)(1+g)} + \frac{(1-\delta)\tilde{k}_t}{(1+n)(1+g)} - \frac{\tilde{k}_t(1+n)(1+g)}{(1+n)(1+g)},$$

which is equivalent to:

$$\Delta \tilde{k}_{t+1} = \tilde{k}_{t+1} - \tilde{k}_t = s_K \frac{(\tilde{k}_t)^\alpha (\tilde{h}_t)^\phi}{(1+n)(1+g)} - \frac{(n + \delta + g + ng)\tilde{k}_t}{(1+n)(1+g)}.$$

Similarly,

$$\Delta \tilde{h}_{t+1} = \tilde{h}_{t+1} - \tilde{h}_t = s_H \frac{(\tilde{k}_t)^\alpha (\tilde{h}_t)^\phi}{(1+n)(1+g)} - \frac{(n + \delta + g + ng)\tilde{h}_t}{(1+n)(1+g)}.$$

Steady State

Set $\tilde{k}_{t+1} = \tilde{k}_t = \tilde{k}^*$ and $\tilde{h}_{t+1} = \tilde{h}_t = \tilde{h}^*$. This implies:

$$\tilde{k}_{t+1} - \tilde{k}_t = \tilde{h}_{t+1} - \tilde{h}_t = 0,$$

which in the Solow equations is equivalent to:

$$s_K \frac{(\tilde{k})^\alpha (\tilde{h})^\phi}{(1+n)(1+g)} - \frac{(n+\delta+g+ng)\tilde{k}}{(1+n)(1+g)} = 0,$$
$$s_H \frac{(\tilde{k})^\alpha (\tilde{h})^\phi}{(1+n)(1+g)} - \frac{(n+\delta+g+ng)\tilde{h}}{(1+n)(1+g)} = 0.$$

Steady State

We can re-write these equations as:

$$\tilde{h} = \left(\frac{(n + \delta + g + ng)\tilde{k}^{1-\alpha}}{s_K} \right)^{\frac{1}{\phi}}$$
$$\tilde{h} = \left(\frac{(n + \delta + g + ng)\tilde{k}^{-\alpha}}{s_H} \right)^{\frac{1}{\phi-1}}$$

These are the two lines along which k and h are constant and which we plot in the phase diagram. Solving the two equations gives:

$$\tilde{k}^* = \left(\frac{s_K^{1-\phi} s_H^{\phi}}{n + g + \delta + ng} \right)^{\frac{1}{(1-\alpha-\phi)}},$$
$$\tilde{h}^* = \left(\frac{s_K^{\alpha} s_H^{1-\alpha}}{n + g + \delta + ng} \right)^{\frac{1}{(1-\alpha-\phi)}}$$

Steady State

Having found the steady state value of human and physical capital stock, we can solve for the steady state of the other variables in the model:

$$\tilde{y}^* = \left(\frac{s_K}{n + g + \delta + ng} \right)^{\frac{\alpha}{(1-\alpha-\phi)}} \left(\frac{s_H}{n + g + \delta + ng} \right)^{\frac{\phi}{(1-\alpha-\phi)}},$$

$$\tilde{w}^* = (1 - \alpha) \left(\frac{s_K}{n + g + \delta + ng} \right)^{\frac{\alpha}{(1-\alpha-\phi)}} \left(\frac{s_H}{n + g + \delta + ng} \right)^{\frac{\phi}{(1-\alpha-\phi)}},$$

$$r^* = \alpha \left(\frac{s_K}{n + g + \delta + ng} \right)^{\frac{\alpha-1}{(1-\alpha-\phi)}} \left(\frac{s_H}{n + g + \delta + ng} \right)^{\frac{\phi}{(1-\alpha-\phi)}},$$

$$\tilde{c}^* = \left(\frac{s_K}{n + g + \delta + ng} \right)^{\frac{\alpha}{(1-\alpha-\phi)}} \left(\frac{s_H}{n + g + \delta + ng} \right)^{\frac{\phi}{(1-\alpha-\phi)}} (1 - s_K - s_H),$$

Variables in per 'Worker' Terms

$$A_t = (1 + g)^t A_0$$

$$k_t = \tilde{k}_t A_t$$

$$\ln(y_t) = \ln(\tilde{y}_t) + \ln(A_t)$$

$$\ln(c_t) = \ln(\tilde{c}_t) + \ln(A_t)$$

$$g_t^y = \ln(y_t) - \ln(y_{t-1})$$

Golden Rule

The growth path for consumption per worker in the steady state is:

$$\begin{aligned} c_t^* &= \tilde{c}^* A_t \\ &= \left(\frac{s_K}{n + g + \delta + ng} \right)^{\frac{\alpha}{(1-\alpha-\phi)}} \left(\frac{s_H}{n + g + \delta + ng} \right)^{\frac{\phi}{(1-\alpha-\phi)}} (1 - s_K - s_H) A_t \end{aligned}$$

Since the natural logarithm is an increasing function, we may as well maximize the altitude of the path for $\ln(c_t^*)$:

$$\begin{aligned} \ln(c_t^*) &= \ln(\tilde{c}^*) + \ln(A_t) \\ &= \frac{\alpha}{(1-\alpha-\phi)} [\ln(s_K) - \ln(n + g + \delta + ng)] + \\ &\quad \frac{\phi}{(1-\alpha-\phi)} [\ln(s_H) - \ln(n + g + \delta + ng)] + \ln(1 - s_K - s_H) + \ln(A_t) \end{aligned}$$

Golden Rule

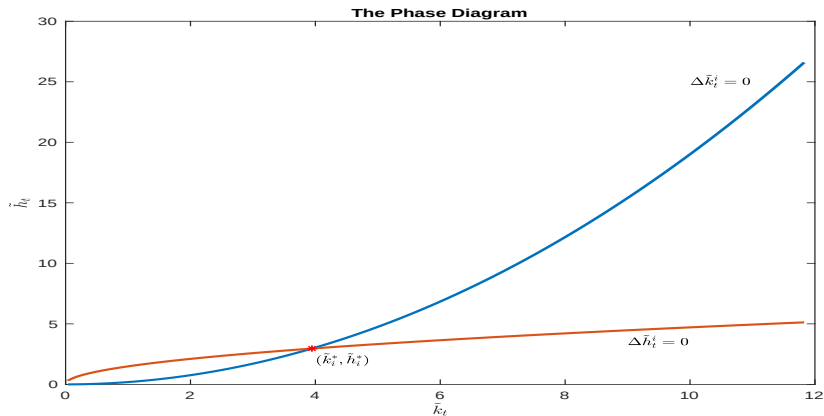
The first-order conditions are:

$$\begin{aligned}\frac{\partial \ln(c_t^*)}{\partial s_K} &= \frac{\alpha}{(1 - \alpha - \phi)} \frac{1}{s_K} - \frac{1}{1 - s_K - s_H} = 0 \\ \Leftrightarrow (1 - \alpha - \phi)s_K &= \alpha(1 - s_K - s_H) \\ \frac{\partial \ln(c_t^*)}{\partial s_H} &= \frac{\phi}{(1 - \alpha - \phi)} \frac{1}{s_H} - \frac{1}{1 - s_K - s_H} = 0 \\ \Leftrightarrow (1 - \alpha - \phi)s_H &= \phi(1 - s_K - s_H)\end{aligned}$$

Solving this system of equations with two unknowns gives $s_K = \alpha$ and $s_H = \phi$.

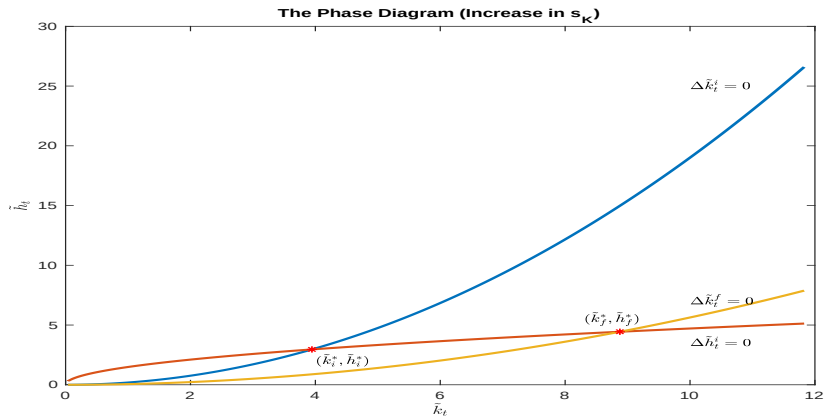
Phase Diagram

Calibration (Lecture 3a, slide 24): $\alpha = \phi = \frac{1}{3}$, $n = 0$, $\delta = 0.06$,
 $g = 0.015$, $s_K^i = 0.2$, $s_H^i = 0.15$.



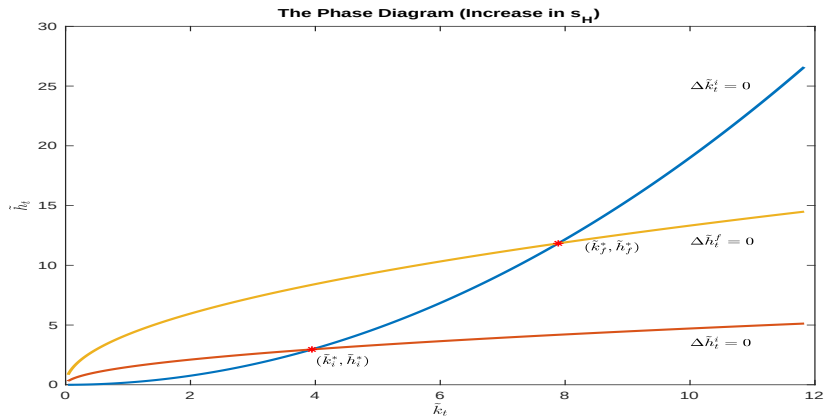
Phase Diagram

Calibration: $\alpha = \phi = \frac{1}{3}$, $n = 0$, $\delta = 0.06$, $g = 0.015$, $s_K^i = 0.2$, $s_K^f = 0.3$, $s_H^i = 0.15$.



Phase Diagram

Calibration: $\alpha = \phi = \frac{1}{3}$, $n = 0$, $\delta = 0.06$, $g = 0.015$, $s_K^i = 0.2$, $s_H^i = 0.15$, $s_H^f = 0.3$.



Comment

In the general Solow model: only one factor can be accumulated;
share $\alpha = \frac{1}{3}$.

Here: 2 factors, with combined share $\alpha + \phi = \frac{2}{3}$.

Exercise 3.1: Simulations

Calibration: $\alpha = \frac{1}{4}$, $\phi = \frac{1}{6}$, $n = 0.01$, $\delta = 0.1$, $g = 0.02$, $s_K = 0.25$, $s_H = 0.25$.

- ▶ $\tilde{k}^*$, $\tilde{y}^*$ increase if either s_K or s_H increases.
- ▶ The effect of the policy on $\tilde{c}^*$ depends on where s_K and s_H were initially and where they end up relative to the Golden Rule.

Increase in the Savings Rate

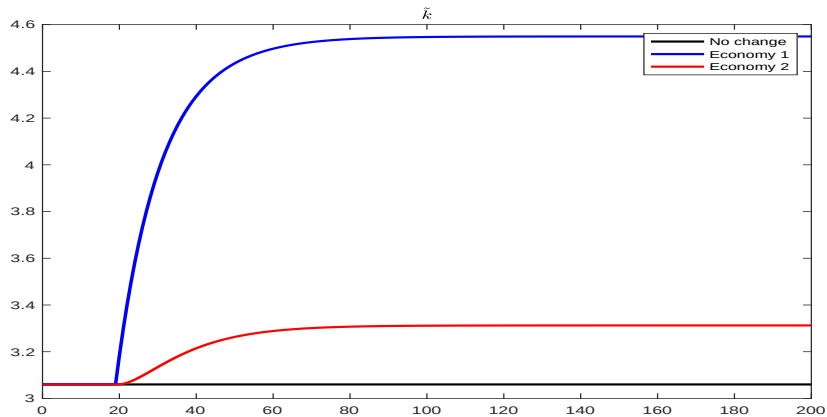
Economy 1:

What happens if the savings rate s_K increases from 0.25 to 0.33?

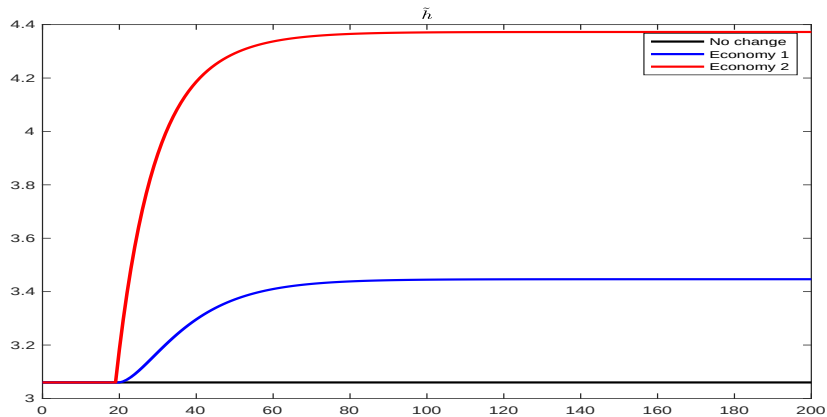
Economy 2:

What happens if the savings rate s_H increases from 0.25 to 0.33?

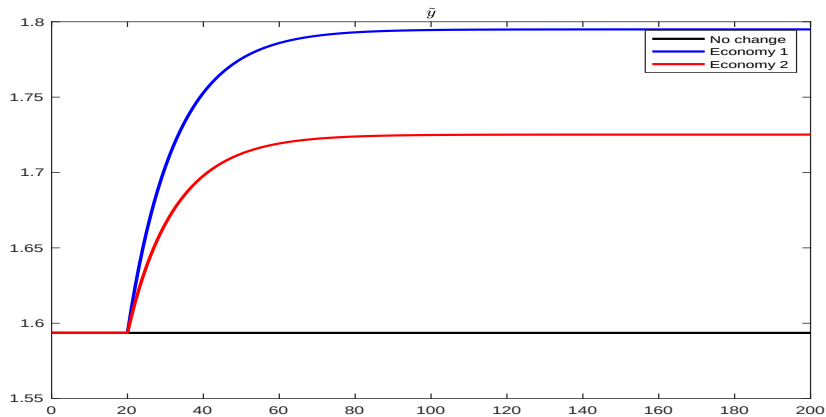
Increase in the Savings Rate



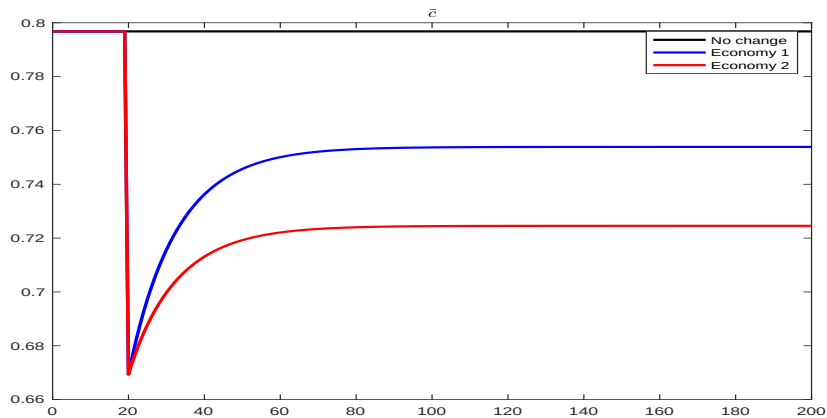
Increase in the Savings Rate



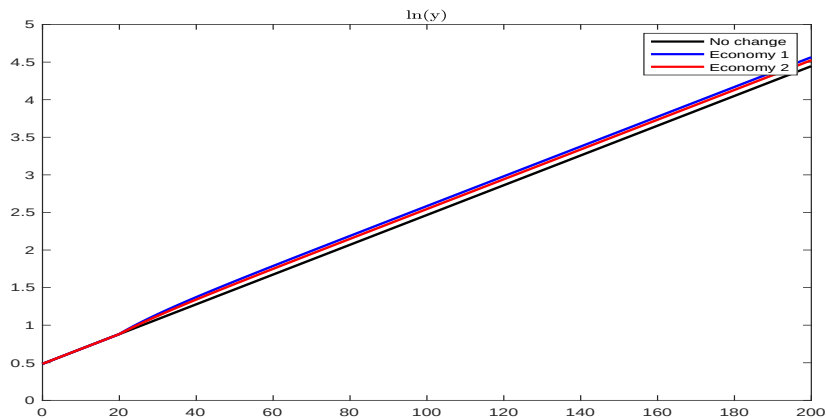
Increase in the Savings Rate



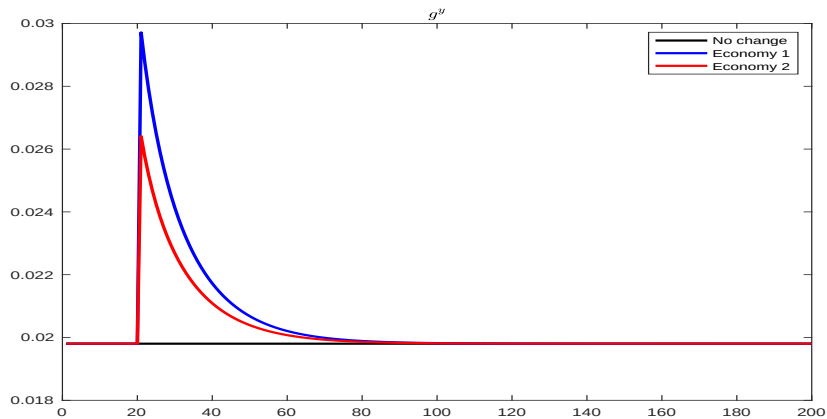
Increase in the Savings Rate



Increase in the Savings Rate



Increase in the Savings Rate



More Realistic Calibration

Calibration: $\alpha = \frac{1}{3}$, $\phi = \frac{1}{3}$, $n = 0.01$, $\delta = 0.1$, $g = 0.02$, $s_K = 0.25$, $s_H = 0.25$.

- ▶ $\tilde{k}^*$, $\tilde{y}^*$ increase if either s_K or s_H increases.
- ▶ The effect of the policy on $\tilde{c}^*$ depends on where s_K and s_H were initially and where they end up relative to the Golden Rule.

Increase in the Savings Rate

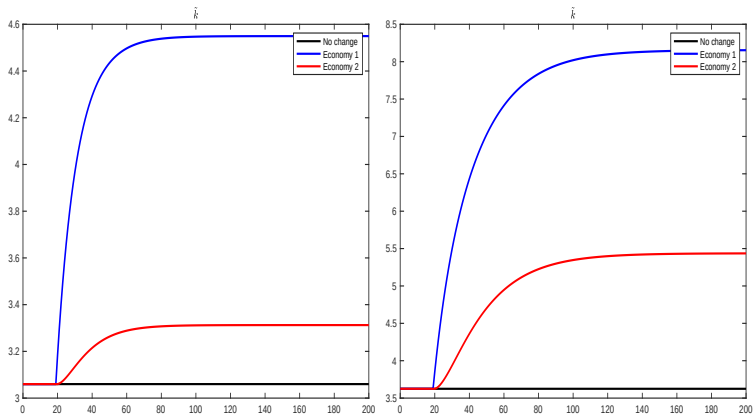
Economy 1:

What happens if the savings rate s_K increases from 0.25 to 0.33?

Economy 2:

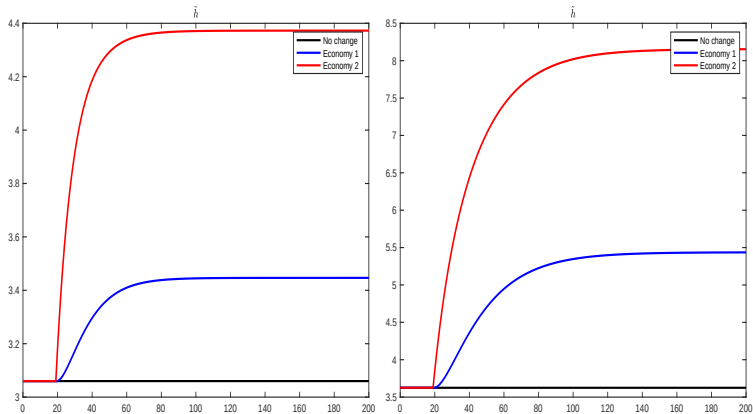
What happens if the savings rate s_H increases from 0.25 to 0.33?

Increase in the Savings Rate



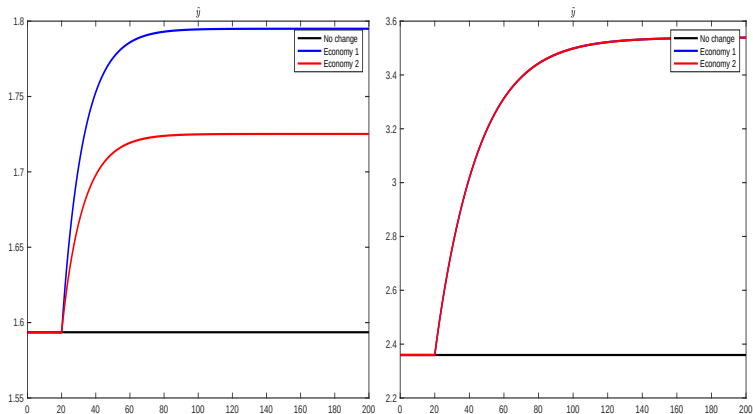
Note: $\alpha = \frac{1}{4}$ & $\phi = \frac{1}{6}$ (left), $\alpha = \phi = \frac{1}{3}$ (right)

Increase in the Savings Rate



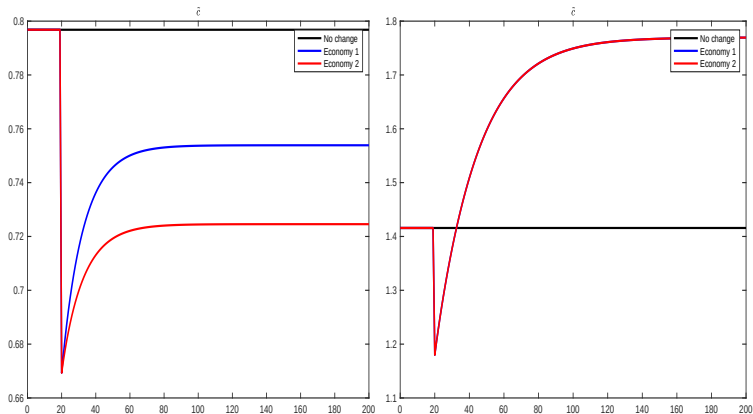
Note: $\alpha = \frac{1}{4}$ & $\phi = \frac{1}{6}$ (left), $\alpha = \phi = \frac{1}{3}$ (right)

Increase in the Savings Rate



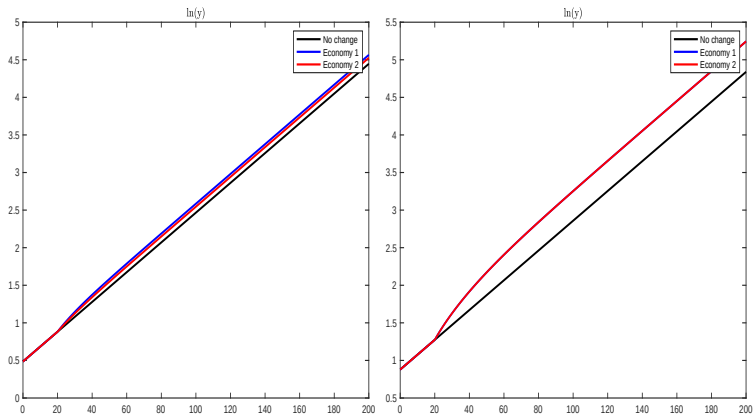
Note: $\alpha = \frac{1}{4}$ & $\phi = \frac{1}{6}$ (left), $\alpha = \phi = \frac{1}{3}$ (right)

Increase in the Savings Rate



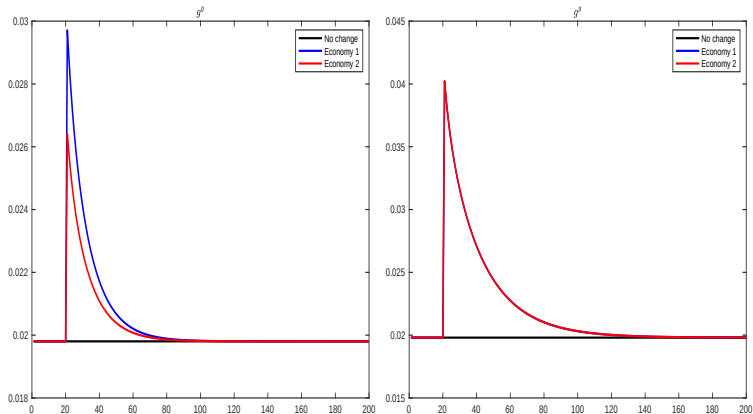
Note: $\alpha = \frac{1}{4}$ & $\phi = \frac{1}{6}$ (left), $\alpha = \phi = \frac{1}{3}$ (right)

Increase in the Savings Rate



Note: $\alpha = \frac{1}{4}$ & $\phi = \frac{1}{6}$ (left), $\alpha = \phi = \frac{1}{3}$ (right)

Increase in the Savings Rate



Note: $\alpha = \frac{1}{4}$ & $\phi = \frac{1}{6}$ (left), $\alpha = \phi = \frac{1}{3}$ (right)

Conclusion about the Savings Rate (Recap)

- ▶ If initially $s < s^{gr}$, then a **small** increase in s results in a long-run increase in consumption in the new steady state.
- ▶ If initially $s > s^{gr}$, then an increase in s results in a long-run decrease in consumption in the new steady state.
- ▶ An economy with $s > s^{gr}$ is “dynamically inefficient” because the economy is saving too much and consumption is being left on the table both in the present and the future.
- ▶ In this case, households would be unambiguously better off by reducing s . Regardless of how impatient people are, a **small** reduction in s gives people more consumption at every date, and this makes them better off.

Exercises 3.2: The Solow Model with Oil

$$Y_t = K_t^\alpha (A_t L_t)^\beta (s_E R_t)^\epsilon, \quad (9)$$

with $\alpha + \beta + \epsilon = 1$. $s_E R_t = E_t$ is energy (oil) and $R_{t+1} = R_t - E_t$ (i.e. $g_t^R = -s_E$).

Problem of the Firm

The optimization problem of the firm is to choose capital, labor, and oil to maximize profits (Π_t):

$$\max_{K_t, L_t, E_t} \Pi_t = F(K_t, L_t, E_t) - w_t L_t - r_t K_t - u_t E_t,$$

- ▶ The first order conditions (FOCs) are:

$$w_t = \frac{\partial Y_t}{\partial L_t} = \beta K_t^\alpha (A_t)^\beta (L_t)^{\beta-1} E_t^\epsilon = \beta K_t^\alpha (A_t)^\beta (L_t)^{\beta-1} (s_E R_t)^\epsilon$$

$$r_t = \frac{\partial Y_t}{\partial K_t} = \alpha K_t^{\alpha-1} (A_t L_t)^\beta E_t^\epsilon = \alpha K_t^{\alpha-1} (A_t L_t)^\beta (s_E R_t)^\epsilon$$

$$u_t = \frac{\partial Y_t}{\partial E_t} = \epsilon K_t^\alpha (A_t L_t)^\beta E_t^{\epsilon-1} = \epsilon K_t^\alpha (A_t L_t)^\beta (s_E R_t)^{\epsilon-1}$$

- ▶ FOCs imply that the firm ought to hire capital, oil, and labor to the point at which the marginal benefit of doing so (the marginal product of capital, oil, or labor) equals the marginal cost of doing so (the factor price).

The Income Shares

The income shares are:

$$\begin{aligned}\frac{r_t K_t}{Y_t} &= \alpha K_t^{\alpha-1} (A_t L_t)^\beta (s_E R_t)^\epsilon \frac{K_t}{Y_t} \\ &= \alpha K_t^{\alpha-1} (A_t L_t)^\beta (s_E R_t)^\epsilon \frac{K_t}{K_t^\alpha (A_t L_t)^\beta (s_E R_t)^\epsilon} = \alpha\end{aligned}$$

$$\begin{aligned}\frac{w_t L_t}{Y_t} &= \beta K_t^\alpha (A_t)^\beta (L_t)^{\beta-1} (s_E R_t)^\epsilon \frac{L_t}{Y_t} \\ &= \beta K_t^\alpha (A_t)^\beta (L_t)^{\beta-1} (s_E R_t)^\epsilon \frac{L_t}{K_t^\alpha (A_t L_t)^\beta (s_E R_t)^\epsilon} = \beta\end{aligned}$$

$$\begin{aligned}\frac{u_t E_t}{Y_t} &= \epsilon K_t^\alpha (A_t L_t)^\beta (s_E R_t)^{\epsilon-1} \frac{E_t}{Y_t} \\ &= \epsilon K_t^\alpha (A_t L_t)^\beta (s_E R_t)^{\epsilon-1} \frac{E_t}{K_t^\alpha (A_t L_t)^\beta (s_E R_t)^\epsilon} = \epsilon\end{aligned}$$

Growth Rates

The production function in per capita units

$$y_t = \frac{Y_t}{L_t} = \frac{K_t^\alpha (A_t L_t)^\beta (E_t)^\epsilon}{L_t^{\alpha+\beta+\epsilon}} = k_t^\alpha (A_t)^\beta (e_t)^\epsilon$$

Take logs:

$$\ln y_t = \alpha \ln k_t + \beta \ln A_t + \epsilon \ln e_t$$

First difference:

$$\ln y_{t+1} - \ln y_t = \alpha (\ln k_{t+1} - \ln k_t) + \beta (\ln A_{t+1} - \ln A_t) + \epsilon (\ln e_{t+1} - \ln e_t)$$

which is equivalent to $g^y = \alpha(g^k) + \beta(g^A) - \epsilon n - \epsilon s_E$. Note that $g^y = g^k$, thus:

$$g^y = \frac{\beta}{1-\alpha} g^A - \frac{\epsilon}{1-\alpha} n - \frac{\epsilon}{1-\alpha} s_E$$

Use that $\alpha + \beta + \epsilon = 1$ to get:

$$g^y = \frac{\beta}{\beta + \epsilon} g^A - \frac{\epsilon}{\beta + \epsilon} n - \frac{\epsilon}{\beta + \epsilon} s_E$$

Growth Rates

We know that $K_t = k_t L_t$ and $Y_t = y_t L_t$. Thus: $g^K = g^Y = g^y + n$, which implies:

$$g^Y = g^K = \frac{\beta}{\beta + \epsilon}(g^A + n) - \frac{\epsilon}{\beta + \epsilon}s_E$$

To show that the growth rate of r_t is zero use $r_t = \alpha \frac{Y_t}{K_t}$. Since $g^Y = g^K$, then $g^r = g^Y - g^K = 0$.

The growth rate of w_t :

$$w_t = \beta \frac{Y_t}{L_t}$$

which implies that $g^w = g^y$. The growth rate of u_t :

$$u_t = \epsilon \frac{Y_t}{s_E R_t}$$

which implies that:

$$g^u = g^Y - g^R = \frac{\beta}{\beta + \epsilon}(g^A + n) - \frac{\epsilon}{\beta + \epsilon}s_E + s_E = \frac{\beta}{\beta + \epsilon}(g^A + n + s_E)$$

Growth Rates

- ▶ Whenever $n > 0$ or $s_E > 0$, one has $g^u > g^w$, so u_t will grow faster than w_t in steady state.
- ▶ Oil becomes increasingly scarce relative to effective labor.
- ▶ This tends (other things being equal) to create a lower marginal product of labor and a higher marginal product of oil, explaining how the factor prices change relative to each other.